

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 2)

The number of solutions of the equation $4 \sin^2 x - 4 \cos^3 x + 9 - 4 \cos x = 0$; $x \in [-2\pi, 2\pi]$ is :

- (1) 1
- (2) 3
- (3) 2
- (4) 0

Q2 - 2024 (27 Jan Shift 1)

Let the set of all $a \in R$ such that the equation $\cos 2x + a \sin x = 2a - 7$ has a solution be $[p, q]$ and

$r = \tan 9^\circ - \tan 27^\circ - \frac{1}{\cot 63^\circ} + \tan 81^\circ$, then pqr is equal to _____

Q3 - 2024 (27 Jan Shift 2)

If $2 \tan^2 \theta - 5 \sec \theta = 1$ has exactly 7 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$, for the least value of $n \in N$ then

$\sum_{k=1}^n \frac{k}{2^k}$ is equal to :

- (1) $\frac{1}{2^{15}}(2^{14} - 14)$
- (2) $\frac{1}{2^{14}}(2^{15} - 15)$
- (3) $1 - \frac{15}{2^{13}}$
- (4) $\frac{1}{2^{13}}(2^{14} - 15)$

Q4 - 2024 (29 Jan Shift 1)

If $\alpha, -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ is the solution of $4 \cos \theta + 5 \sin \theta = 1$, then the value of $\tan \alpha$ is

- (1) $\frac{10 - \sqrt{10}}{6}$
- (2) $\frac{10 - \sqrt{10}}{12}$
- (3) $\frac{\sqrt{10} - 10}{12}$
- (4) $\frac{\sqrt{10} - 10}{6}$

Q5 - 2024 (29 Jan Shift 2)

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Questions with Answer Keys

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The sum of the solutions $x \in \mathbb{R}$ of the equation $\frac{3 \cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = x^3 - x^2 + 6$ is

- (1) 0
- (2) 1
- (3) -1
- (4) 3

Q6 - 2024 (30 Jan Shift 1)

If $2 \sin^3 x + \sin 2x \cos x + 4 \sin x - 4 = 0$ has exactly 3 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$, $n \in \mathbb{N}$, then the roots of the equation $x^2 + nx + (n - 3) = 0$ belong to :

- (1) $(0, \infty)$
- (2) $(-\infty, 0)$
- (3) $\left(-\frac{\sqrt{17}}{2}, \frac{\sqrt{17}}{2}\right)$
- (4) $\mathbb{Z}$

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Questions with Answer Keys

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Solutions

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Q1

$$4 \sin^2 x - 4 \cos^3 x + 9 - 4 \cos x = 0; x \in [-2\pi, 2\pi]$$

$$4 - 4 \cos^2 x - 4 \cos^3 x + 9 - 4 \cos x = 0$$

$$4 \cos^3 x + 4 \cos^2 x + 4 \cos x - 13 = 0$$

$$4 \cos^3 x + 4 \cos^2 x + 4 \cos x = 13$$

L.H.S. ≤ 12 can't be equal to 13.

Q2

$$\cos 2x + a \cdot \sin x = 2a - 7$$

$$a(\sin x - 2) = 2(\sin x - 2)(\sin x + 2)$$

$$\sin x = 2, \quad a = 2(\sin x + 2)$$

$$\Rightarrow a \in [2, 6]$$

$$p = 2 \quad q = 6$$

$$r = \tan 9^\circ + \cot 9^\circ - \tan 27^\circ - \cot 27^\circ$$

$$r = \frac{1}{\sin 9^\circ \cdot \cos 9^\circ} - \frac{1}{\sin 27^\circ \cdot \cos 27^\circ}$$

$$= 2 \left[\frac{4}{\sqrt{5} - 1} - \frac{4}{\sqrt{5} + 1} \right]$$

$$r = 4$$

$$p \cdot q \cdot r = 2 \times 6 \times 4 = 48$$

Q3

$$2 \tan^2 \theta - 5 \sec \theta - 1 = 0$$

$$\Rightarrow 2 \sec^2 \theta - 5 \sec \theta - 3 = 0$$

$$\Rightarrow (2 \sec \theta + 1)(\sec \theta - 3) = 0$$

$$\Rightarrow \sec \theta = -\frac{1}{2}, 3$$

$$\Rightarrow \cos \theta = -2, \frac{1}{3}$$

$$\Rightarrow \cos \theta = \frac{1}{3}$$

For 7 solutions $n = 13$

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Solutions

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$$\text{So, } \sum_{k=1}^{13} \frac{k}{2^k} = S \text{ (say)}$$

$$S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{13}{2^{13}}$$

$$\frac{1}{2} S = \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{12}{2^{13}} + \frac{13}{2^{14}}$$

$$\Rightarrow \frac{S}{2} = \frac{1}{2} \cdot \frac{1 - \frac{1}{2^{13}}}{1 - \frac{1}{2}} - \frac{13}{2^{14}} \Rightarrow S = 2 \cdot \left(\frac{2^{13} - 1}{2^{13}} \right) - \frac{13}{2^{13}}$$

Q4

$$4 + 5 \tan \theta = \sec \theta$$

$$\text{Squaring : } 24 \tan^2 \theta + 40 \tan \theta + 15 = 0$$

$$\tan \theta = \frac{-10 \pm \sqrt{10}}{12}$$

$$\text{and } \tan \theta = -\left(\frac{10 + \sqrt{10}}{12}\right) \text{ is Rejected.}$$

(3) is correct.

Q5

$$\frac{3 \cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = x^3 - x^2 + 6$$

$$\Rightarrow \frac{\cos 2x (3 + \cos^2 2x)}{\cos 2x (1 - \sin^2 x \cos^2 x)} = x^3 - x^2 + 6$$

$$\Rightarrow \frac{4 (3 + \cos^2 2x)}{(4 - \sin^2 2x)} = x^3 - x^2 + 6$$

$$\Rightarrow \frac{4 (3 + \cos^2 2x)}{(3 + \cos^2 2x)} = x^3 - x^2 + 6$$

$$x^3 - x^2 + 2 = 0 \Rightarrow (x + 1)(x^2 - 2x + 2) = 0$$

so, sum of real solutions = -1

Q6

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Solutions

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$$2 \sin^3 x + 2 \sin x \cdot \cos^2 x + 4 \sin x - 4 = 0$$

$$2 \sin^3 x + 2 \sin x \cdot (1 - \sin^2 x) + 4 \sin x - 4 = 0$$

$$6 \sin x - 4 = 0$$

$$\sin x = \frac{2}{3}$$

 $n = 5$ (in the given interval)

$$x^2 + 5x + 2 = 0$$

$$x = \frac{-5 \pm \sqrt{17}}{2}$$

$$x = \frac{2}{-5 \pm \sqrt{17}}$$

Required interval $(-\infty, 0)$

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